

RAMU LAKSHMANAN

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Profile

Year 4 Computer Science undergraduate with a demonstrated passion for AI and Biology. Distinguished by strong analytical skills honed through self-initiated research projects. Skilled communicator adept at presenting complex ideas to diverse audiences. US Citizen.

Technical Skills

Languages/Software: Python, R, Julia, Bash, Java, Javascript, PDDL, VMD, Gromacs, Snakemake

Packages/Frameworks: Pandas, Matplotlib, Sklearn, Pytorch, ReactJS, MaterialUI

Education

National University of Singapore

Aug 2022 – Jul 2026

B.Comp in Computer Science, Minor in History

- * Awarded the following honours: NUS Outstanding Undergraduate Researcher Prize, NUS Innovation & Research Award
- * GPA: 4.52/5.00 (First Class Honours)

Research Experience

RESEARCH STUDENT - MTMS, Genome Institute of Singapore, A*STAR

Aug 2025 – May 2026

MTMS is a lab that specialises in studying microbial communities

- * Benchmarked attribution-based explainability methods over synthetic genomic sequence datasets, demonstrating their systemic inability to reflect compositional rules. [Published in LMRL@ICLR](#)
- * Implemented [CORAL](#), a gradient-based sequence optimiser for genomic foundation models to generate rule-aligned counterfactual sequences

ML & COMPUTATIONAL BIOLOGY INTERN - Hummingbird Bioscience

May 2024 – Dec 2024

Hummingbird Bioscience is a clinical-stage biotech startup in Singapore that specialises in developing antibodies against hard-to-target cancer targets.

- * Designed a novel and validated directed evolution pipeline to generate nanobody lead candidates using generative protein language models screened using structure prediction methods
- * Engineered an XGBoost Model to rank the relative hydrophobicity of antibody-drug conjugates in a data-limited setting, outperforming existing models
- * Optimised in-house antibody humanisation pipelines using GPU acceleration and parallelisation, resulting in a 40x speedup

WIKI LEAD - Team NUS iGEM

Feb 2023 – Nov 2023

iGEM is an international genetic engineering competition where a multidisciplinary team develops innovative solutions to tackle pressing challenges through Synthetic Biology

- * Created 2 AI language models using Pytorch to predict generalised RNA - RNA interactions and RNA switch performance respectively
- * Led the development of 2 websites: One to document the project and another for members of the Synthetic Biology community to deploy and host ML models
- * Achieved 3rd in the Undergraduate Category out of 167 teams, nominated for the Best Model & Best Wiki awards

Projects

Inverse Planning under Domain Uncertainty | <https://github.com/laksherthanlife/InversePlanning.jl>

Apr 2026

- * Implemented a SAT-based static domain learner to infer PDDL representations from action traces alone using Julia
- * Extended a SOTA inverse planning algorithm, SIPS, to infer agent's perception of state, in addition to goal inference

Measuring Genetic Privacy in cBioPortal | <https://github.com/laksherthanlife/Genomic-Privacy>

Nov 2025

- * Performed a Membership Inference attack on cBioPortal's breast cancer datasets using likelihood ratio tests, demonstrating a dataset identification rate of 34% TPR for individuals at 1% FPR
- * Analysed K-anonymity of cBioPortal's breast cancer datasets and implemented a sparse set-cover approach to find the minimum mutations to identify patients, determining 4 mutations were sufficient to fingerprint 50% of entries

Cancer Ensemble Variant Caller | <https://github.com/laksherthanlife/EnsembleVariantCaller>

Feb 2024

- * Created a Random Forest Model to ensemble 4 Cancer Variant Calling models as a part of a graduate level coursework, achieving the top performance in class
- * Developed a VCF parser in R to aggregate features from each variant caller for model training